

URALSK, Kazakhstan

WMO: 351080

Lat: 51.151N

Lon: 51.543E

Elev: 38

StdP: 100.87

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.0	-25.9	-31.9	0.2	-28.8	-28.9	0.3	-25.7	12.1	-5.6	10.9	-6.7	2.0	0	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	35.9	19.7	33.9	19.2	32.0	18.7	21.4	30.9	20.6	30.1	19.8	29.1	3.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.6	23.8	17.8	12.8	23.3	16.8	12.0	22.9	62.2	30.8	59.3	30.2	56.8	28.8	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	9.0	8.2	-32.3	38.5	3.8	1.6	-35.1	39.7	-37.3	40.6	-39.4	41.6	-42.1	42.8		
(5)			-32.5	23.5	3.7	1.3	-35.2	24.4	-37.3	25.2	-39.4	25.9	-42.1	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-11.4	-11.2	-3.6	8.2	15.7	20.5	22.7	21.5	14.7	6.8	-1.3	-8.3	(6)
(7)		DBStd	13.65	7.43	7.33	6.12	5.60	4.64	4.41	3.59	4.34	4.70	4.76	6.07	7.15	(7)
(8)		HDD10.0	2829	664	594	422	98	7	1	0	0	11	123	339	569	(8)
(9)		HDD18.3	4793	923	828	679	305	107	27	6	18	128	357	589	827	(9)
(10)		CDD10.0	1475	0	0	1	44	186	316	395	357	151	25	1	0	(10)
(11)		CDD18.3	397	0	0	0	1	27	92	142	117	18	0	0	0	(11)
(12)		CDH23.3	5404	0	0	0	36	439	1248	1806	1567	304	4	0	0	(12)
(13)		CDH26.7	2459	0	0	0	4	149	582	855	764	104	0	0	0	(13)
(14)	Wind	WSAvg	3.7	4.0	4.1	4.2	4.2	3.6	3.3	3.2	3.2	3.5	3.8	3.9	(14)	
(15)	Precipitation	PrecAvg	367	28	22	27	26	33	35	38	29	29	38	31	29	(15)
(16)		PrecMax	494	63	39	63	48	85	99	97	82	70	87	81	70	(16)
(17)		PrecMin	207	8	7	1	3	1	3	4	4	3	2	3	2	(17)
(18)		PrecStd	77	12	8	17	13	19	20	29	21	18	21	20	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.1	2.2	15.2	27.0	32.4	37.2	37.9	38.1	32.7	23.7	12.9	5.2	(19)
(20)			MCWB	1.5	1.4	8.8	14.9	18.1	19.9	20.2	19.9	16.5	12.9	10.4	4.3	(20)
(21)		2%	DB	1.1	1.2	9.0	23.9	30.2	35.0	35.2	35.8	29.8	20.8	9.9	2.2	(21)
(22)			MCWB	0.7	0.6	5.7	13.6	17.2	19.6	19.8	19.5	15.8	11.6	7.9	1.6	(22)
(23)		5%	DB	0.2	0.6	5.5	21.1	28.2	32.6	33.2	33.4	27.0	17.6	7.8	1.2	(23)
(24)			MCWB	0.0	0.2	3.5	12.2	16.7	18.9	19.6	18.8	15.0	10.5	6.6	0.8	(24)
(25)		10%	DB	-0.8	-0.4	3.2	18.1	25.8	30.1	31.4	31.1	24.1	14.6	5.9	0.6	(25)
(26)			MCWB	-1.2	-0.8	1.8	10.6	15.7	18.4	19.1	18.5	14.2	9.7	4.8	0.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	1.6	8.9	16.2	20.0	22.5	22.6	21.8	18.7	15.3	11.1	4.6	(27)
(28)			MCDWB	1.8	1.8	14.8	25.1	29.0	31.6	31.3	32.3	28.5	20.3	12.5	5.2	(28)
(29)		2%	WB	0.8	0.8	5.9	14.3	18.6	21.0	21.6	20.7	17.1	12.8	8.2	1.9	(29)
(30)			MCDWB	0.9	1.0	8.5	21.9	27.5	30.5	30.6	31.4	25.2	17.9	9.2	2.1	(30)
(31)		5%	WB	0.0	0.1	3.7	12.8	17.5	20.1	20.7	20.0	16.1	11.5	6.5	0.9	(31)
(32)			MCDWB	0.3	0.5	5.3	19.5	26.4	29.4	29.9	30.4	24.3	16.1	7.5	1.1	(32)
(33)		10%	WB	-1.2	-1.0	1.9	11.2	16.2	19.2	20.0	19.1	15.0	10.1	4.7	0.1	(33)
(34)			MCDWB	-0.8	-0.5	2.9	17.4	24.2	28.0	28.8	28.8	22.1	13.7	5.6	0.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	9.3	8.8	12.1	13.4	14.0	13.8	14.6	13.7	10.6	6.9	7.2	(35)
(36)			MCDBR	5.2	6.2	9.7	15.9	17.3	17.3	16.4	17.5	17.9	15.7	8.4	5.3	(36)
(37)			MCWBR	5.2	6.1	6.8	7.9	7.5	6.7	5.8	6.6	8.0	8.8	6.9	5.0	(37)
(38)		5% WB	MCDBR	5.3	6.5	9.3	14.7	15.8	14.8	14.3	15.6	15.4	13.2	7.5	4.8	(38)
(39)			MCWBR	5.4	6.4	6.8	8.0	7.7	6.5	5.6	6.5	7.9	8.4	6.5	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.314	0.337	0.386	0.439	0.417	0.408	0.414	0.429	0.383	0.358	0.337	0.312	(40)	
(41)		taud	2.057	2.013	2.010	2.070	2.222	2.308	2.297	2.227	2.317	2.312	2.239	2.108	(41)	
(42)		Ebn at Noon	656	754	789	796	838	849	836	799	798	741	642	595	(42)	
(43)		Edh at Noon	83	114	139	148	133	123	123	125	103	85	70	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.07	2.23	3.47	4.56	5.94	6.49	6.32	5.39	3.75	2.06	0.93	0.68	(44)	
(45)		RadStd	0.26	0.24	0.52	0.43	0.42	0.54	0.44	0.37	0.44	0.28	0.12	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page